E-COMMERCE & APPLICATIONS – UNIT I

1. Understanding E-Commerce

A. Definition and Evolution

- **Electronic Commerce (E-Commerce)** refers to the digitally enabled commercial transactions between and among organizations and individuals.
- **Scope:** It digitally enables transactions involving the exchange of value (such as money, services, or information) across organizational or individual boundaries using information and communications technology (ICT), primarily the Internet and the World Wide Web.
- **Evolution:**
    - **Phase 1: Innovation (1995–2000):** Rapid commercialization, dot-com boom, focus on building infrastructure and storefronts.
    - **Phase 2: Consolidation (2001–2006):** Post-bubble correction, shift toward business profitability, brand reinforcement, and multi-channel strategies.
    - **Phase 3: Reinvention (2007–Present):** Rise of social media, mobile platforms (m-commerce), local marketing, cloud computing, and on-demand service platforms.

B. Traditional Commerce vs. E-Commerce

Feature	Traditional Commerce	E-Commerce
Market Reach	Local or regional; constrained by physical presence.	Global; operates across geographical boundaries 24/7.
Information Richness	High (face-to-face interaction, personal selling).	High digital richness (multimedia, detailed specifications, user reviews).
Transaction Cost	High (intermediaries, physical store overheads, labor).	Low (direct-to-consumer disintermediation, automated processing).
Search Cost	High (requires physical travel and manual comparison).	Low (instant price comparison engines and search algorithms).

C. Unique Features of E-Commerce Technology

1. **Ubiquity:** Available everywhere, at all times, liberating the market from physical space and time constraints.
2. **Global Reach:** Reaches across national boundaries seamlessly.
3. **Universal Standards:** Shared technical standards across nations reduce market entry costs and search friction.

4. **Information Richness:** Supports video, audio, and text messages simultaneously, delivering complex information to large audiences.
5. **Interactivity:** Allows two-way communication between merchant and consumer, acting like an interactive online sales representative.
6. **Information Density:** Vastly increases information abundance while lowering storage, processing, and communication costs.
7. **Personalization and Customization:** Enables merchants to target marketing messages to specific individuals (personalization) and modify products/services based on user preference (customization).

2. E-Commerce Business Models and Concepts

- **Business Model:** A set of planned activities (sometimes called business processes) designed to result in a profitable profit in a marketplace. It outlines how a company produces, delivers, and captures value.

Key Elements of a Business Model:

1. **Value Proposition:** Why should the customer buy from this firm? (e.g., personalization, convenience, price).
2. **Revenue Model:** How does the firm plan to earn money? (e.g., advertising, subscription, transaction fee, sales, affiliate).
3. **Market Opportunity:** What is the market space the company intends to serve, and what is its revenue potential (Total Addressable Market - TAM)?
4. **Competitive Environment:** Who are the other players in the market space, and what are their market shares and strengths?
5. **Competitive Advantage:** What special advantages does the firm bring to the market space (e.g., first-mover advantage, asymmetric information, superior products)?
6. **Market Strategy:** The plan detailing how the company will promote its products to attract customers.
7. **Organizational Development:** The structural framework outlining how the company will organize the work needed to accomplish its business plan.
8. **Management Team:** The employees who have the background and experience to make the business model work.

3. Major Business-to-Consumer (B2C) Business Models

Business-to-Consumer (B2C) e-commerce involves businesses selling products or services directly to individual consumers.

B2C Model Type	Description & Mechanism	Typical Examples
E-Tailer	Online retail store; sells physical goods directly to consumers or businesses.	Amazon, Flipkart, Myntra
Community Provider	Creates a digital online environment where people with similar interests can transact and communicate.	Facebook, Reddit, LinkedIn

Content Provider	Distributes information content (news, music, video, artwork) digitally over the web.	Netflix, New York Times, Spotify
Portal	Offers initial entry point to the web along with specialized content and services (search engines, news, email).	Yahoo, MSN, Google
Transaction Broker	Processes user transactions for consumers, primarily handling actions such as stock trading, travel bookings, or job matching.	E-Trade, MakeMyTrip, Expedia
Market Creator	Builds a digital environment where buyers and sellers can meet, display products, search for products, and establish prices.	eBay, Airbnb, Uber
Service Provider	Offers services online rather than physical products (e.g., web hosting, photo sharing, online banking).	Dropbox, Canva, TurboTax

4. Major Business-to-Business (B2B) Business Models

Business-to-Business (B2B) e-commerce involves commercial transactions between corporate entities, representing the largest sector of e-commerce by volume.

Key B2B Classifications:

1. **Net Marketplaces:** Digital marketplaces that bring hundreds of suppliers and buyers together in a single platform.
    - **E-Distributors:** Supply products directly to individual businesses (e.g., Grainger, Alibaba industrial supplies).
    - **Exchanges:** Independent digital marketplaces where suppliers and commercial buyers conduct spot purchasing in open, competitive price environments.
    - **Industrial Consortia:** Industry-owned vertical marketplaces that enable buyers to purchase from specialized suppliers.
2. **Private Industrial Networks (Private Trading Exchanges - PTX):** Web-enabled networks designed to coordinate communication and business processes between a firm and its chosen business partners (e.g., Walmart's supply chain network connecting directly to its trusted manufacturers).

5. Business Models in Emerging E-Commerce Areas

1. **Social E-Commerce:** E-commerce enabled by social networks and online social relationships (e.g., Facebook Marketplace, Instagram Shopping, social commerce recommendations).
2. **Mobile E-Commerce (M-Commerce):** E-commerce transactions and services executed via wireless mobile devices like smartphones and tablets.

3. **Local E-Commerce:** A form of e-commerce that is focused on engaging the consumer based on their current geographic location (e.g., Zomato, Swiggy, local Google search ads).
4. **On-Demand Economy / Gig Economy:** Platforms that connect consumer demand for immediate services with available providers using digital apps (e.g., Uber, TaskRabbit).

6. How the Internet and the Web Change Business

The integration of the Internet and the Web has structurally transformed the business landscape across multiple dimensions:

- **Reduction of Information Asymmetry:** Buyers now possess nearly equal access to pricing, product quality, and competitor data, shifting market power toward consumers.
- **Disintermediation and Re-intermediation:**
    - *Disintermediation:* Elimination of traditional middlemen (wholesalers, distributors) as manufacturers sell directly to end consumers.
    - *Re-intermediation:* Emergence of new digital intermediaries (like aggregators, comparison engines, and digital platform operators) that provide value through matching and filtering.
- **Network Effects:** As more users join a network, the utility and value of the network for each participant increases exponentially (e.g., marketplaces become more valuable as more buyers and sellers join).
- **Speed and Agility:** Real-time inventory tracking, automated customer relationship management (CRM), and global supply chain visibility enable firms to operate with leaner inventories and faster response times.

E-COMMERCE & APPLICATIONS – UNIT II

PART 1: E-COMMERCE INFRASTRUCTURE

1. The Internet: Technology Background

- **Deep Definition:** The Internet is a worldwide system of interconnected computer networks that use the standard Internet Protocol Suite (TCP/IP) to link billions of devices globally.
- **Core Technological Pillars:**
    1. **Packet Switching:** A method of slicing digital messages into smaller parcels called **packets**, routing them independently across various network paths, and reassembling them at their destination. This eliminates the need for a dedicated physical circuit for each connection.
    2. **Transmission Control Protocol / Internet Protocol (TCP/IP):**
        - **IP (Internet Protocol):** Handles the addressing and routing of packets across networks.
        - **TCP (Transmission Control Protocol):** Establishes connections, tracks packet sequences, and manages error detection and retransmission if packets are lost.
    3. **Client-Server Architecture:** A computing model where powerful computers (**Servers**) store and serve web pages, data, and applications to desktop or mobile devices (**Clients**) upon request.

2. The Internet Today

- **Next-Generation Characteristics:**
    - **Broadband & High-Speed Access:** Transition from dial-up to high-speed fiber-optic, 4G/5G, and satellite internet, enabling high-definition streaming and instantaneous commercial transactions.

- **Cloud Computing:** Shifting infrastructure from local hardware to scalable cloud providers (AWS, Microsoft Azure, Google Cloud), allowing e-commerce sites to scale resources dynamically during peak traffic (e.g., Black Friday sales).
- **Internet of Things (IoT) Integration:** Connected devices (smart home appliances, wearables) that automatically reorder supplies or transmit usage data to e-commerce platforms.
- **IPv6 Transition:** Expansion from IPv4 to IPv6 addressing to accommodate the exploding number of connected internet-enabled devices.

3. The World Wide Web (WWW)

- **Deep Definition:** The World Wide Web is an information space where documents and other web resources are identified by Uniform Resource Locators (URLs), linked by hypertext links, and accessible via the Internet. Developed by Tim Berners-Lee in 1989.
- **Core Web Components:**
  - **HyperText Transfer Protocol (HTTP / HTTPS):** The underlying protocol used to format and transmit messages between web browsers and web servers.
  - **HTML (HyperText Markup Language):** The standard markup language used to create and structure web pages and applications.
  - **URLs (Uniform Resource Locators):** The global address of documents and other resources on the World Wide Web (e.g., `[https://www.example.com/store](https://www.example.com/store)`).
  - **Web Browsers & Web Servers:** Browsers (Chrome, Safari, Firefox) act as client software rendering web pages, while web servers (Apache, Nginx) host and deliver those pages.

PART 2: BUILDING AN E-COMMERCE WEBSITE

1. A Systematic Approach (SDLC for E-Commerce)

Building a successful e-commerce website requires a structured systems development life cycle (SDLC) divided into sequential phases:

1. **Systems Analysis / Planning:**
   - Identify business objectives, system functionality requirements, and information system capabilities needed.
2. **System Design:**
   - **Logical Design:** Data flow diagrams, processing logic, and security architecture.
   - **Physical Design:** Hardware specifications, software selection, database schemas, and telecommunications paths.
3. **Building the System:**
   - Choosing whether to **outsource** development, build **in-house**, or use pre-built **e-commerce platforms / SaaS builders** (like Shopify or WooCommerce).
4. **Testing:**
   - Comprehensive testing including unit testing, system testing, load testing (simulating heavy traffic spikes), and security vulnerability testing.
5. **Implementation & Maintenance:**
   - Deploying the site to a live production server, monitoring uptime, tracking user analytics, and updating software regularly.

2. Choosing Server Software

- **Deep Definition:** Web server software is responsible for locating and delivering requested web pages, managing HTTP requests, and maintaining security layers.
- **Categories of E-Commerce Software:**

1. **Web Server Operating Systems & Daemons:**
    - **Apache HTTP Server & Nginx:** Open-source web servers that handle incoming HTTP requests and serve static/dynamic web content.
2. **E-Commerce Platform Software Packages:**
    - **Open-Source / Self-Hosted:** *WooCommerce* (WordPress plugin), *Magento (Adobe Commerce)*. Offers full code control and customization, but requires technical maintenance and hosting management.
    - **SaaS (Software as a Service):** *Shopify, BigCommerce*. Hosted solutions providing templates, secure payment processing, and hosting bundled into a monthly subscription. Highly reliable and easy to deploy.

3. Choosing Hardware for an E-Commerce Site

- **Key Hardware Requirements:**
    - **Processing Power (CPU):** High-frequency multi-core processors to handle concurrent database queries, session management, and dynamic page rendering.
    - **Random Access Memory (RAM):** Sufficient memory to cache catalogs, user session data, and database indexes, preventing disk bottlenecks.
    - **Storage (SSD / NVMe):** High-speed solid-state drives for rapid database reads/writes and product asset delivery.
- **Scalability & Load Balancing:**
    - For high-traffic sites, a single server is insufficient. **Load balancers** distribute incoming user traffic across a cluster of multiple web servers.
    - **Cloud Infrastructure:** Utilizing cloud instances allows elastic scaling—automatically provisioning more virtual servers during high-traffic events and scaling down when traffic subsides.

4. Other E-Commerce Site Tools

To function efficiently, an e-commerce website integrates several vital auxiliary tools:

1. **Digital Payment Gateways:** Secure third-party processors (e.g., Stripe, PayPal, Razorpay) that authorize credit card, UPI, and digital wallet transactions.
2. **SSL/TLS Certificates:** Secure Sockets Layer / Transport Layer Security protocols that encrypt data transmitted between the customer's browser and the web server, represented by the padlock icon (`https://`).
3. **Web Analytics Tools:** Platforms (e.g., Google Analytics) that track user behavior, conversion rates, bounce rates, and traffic sources.
4. **Customer Relationship Management (CRM) & Inventory Tools:** Software that syncs online orders with warehouse inventory levels and manages customer support tickets and email marketing campaigns.

E-COMMERCE & APPLICATIONS – UNIT III

1. The E-Commerce Security Environment

A. Dimensions of E-Commerce Security

To achieve secure e-commerce transactions, a robust infrastructure must satisfy six key dimensions of security:

1. **Integrity:** The ability to ensure that information being displayed, transmitted, or stored has not been altered or tampered with by unauthorized parties.
2. **Non-repudiation:** The ability to ensure that e-commerce participants do not deny (repudiate) their online actions or transactions (e.g., denying that they placed an order or authorized a payment).
3. **Authenticity:** The ability to verify the identity of the person or entity with whom you are transacting online.
4. **Confidentiality:** The ability to ensure that messages and data are available only to those authorized to view them (preventing unauthorized interception).
5. **Privacy:** The ability to control the use of information a customer provides about themselves to an e-commerce merchant.
6. **Availability:** The ability to ensure that an e-commerce site functions continuously and is accessible to users without disruption.

B. The Security vs. Convenience Trade-Off

- Implementing rigorous security measures (multi-factor authentication, biometric verification, heavy encryption layers) often increases friction for the user, potentially lowering conversion rates.
- E-commerce architects must balance maximum security against user-friendly convenience and system performance.

2. Security Threats in the E-Commerce Environment

E-commerce networks face diverse threats originating from internal and external malicious actors:

A. Malicious Code & Software Threats

- **Viruses & Worms:** Self-replicating programs that attach to clean files or spread independently across networks, corrupting data or systems.
- **Trojan Horses:** Programs that appear benign or useful but conceal malicious code designed to steal data or create backdoors.
- **Ransomware:** Malware that encrypts a victim's files or database, demanding financial ransom in exchange for the decryption key.
- **Spyware / Adware:** Software installed covertly to track user browsing habits or display unsolicited advertisements.

B. Network & Infrastructure Attacks

- **Denial of Service (DoS) & Distributed Denial of Service (DDoS):** Flooding an e-commerce server with millions of fraudulent requests from botnets to exhaust system resources, rendering the website inaccessible to legitimate customers.
- **Sniffing:** Monitoring and intercepting data packets as they travel across computer networks using packet-sniffing software.
- **Man-in-the-Middle (MitM) Attack:** An attacker secretly intercepts and relays communications between two parties who believe they are directly communicating with each other, allowing the attacker to steal or modify data.

C. Application & Web-Level Attacks

- **Phishing & Social Engineering:** Deceptive emails, SMS messages, or fake websites mimicking trusted brands to trick users into revealing sensitive login credentials or credit card numbers.
- **SQL Injection (SQLi):** Inserting malicious SQL statements into web form input fields to manipulate backend databases, bypass authentication, or dump customer records.

- **Cross-Site Scripting (XSS):** Injecting malicious client-side scripts into trusted web pages viewed by other users, allowing attackers to hijack sessions or steal cookies.

3. Technology Solutions for E-Commerce Security

A. Cryptography & Encryption

Encryption transforms plain text into unciphered ciphertext using mathematical algorithms and keys, protecting data at rest and in transit.

- **Symmetric Key Encryption:** Uses the *same single key* for both encryption and decryption. Fast, but requires secure distribution of the key beforehand.
- **Asymmetric (Public-Key) Encryption:** Uses a *mathematically related pair of keys*:
    - **Public Key:** Distributed openly to encrypt messages.
    - **Private Key:** Kept secret by the owner to decrypt messages.

B. Secure Sockets Layer (SSL) and Transport Layer Security (TLS)

- **TLS/SSL Protocols:** Standard security technologies used to establish an encrypted link between a web server and a browser.
- **Mechanism:** TLS combines symmetric and asymmetric encryption: asymmetric encryption is used safely to exchange a session key, which is then used for fast symmetric encryption during the browsing session.

C. Perimeter Defenses

- **Firewalls:** Hardware or software systems that filter incoming and outgoing network traffic based on predetermined security rules, blocking unauthorized access.
- **Intrusion Detection Systems (IDS) & Intrusion Prevention Systems (IPS):** Monitoring tools that analyze network traffic for malicious signatures or anomalous behavior and trigger alerts or block traffic.

4. Policies, Procedures, and Laws

Technology alone cannot secure an e-commerce platform; organizational frameworks are equally vital:

- **Security Policies:** Formal written documents defining acceptable use, password complexity rules, data classification, and incident response protocols.
- **Risk Assessment:** Systematic evaluation of organizational assets, vulnerabilities, and potential threats to prioritize security investments.
- **Compliance Standards:**
    - **PCI-DSS (Payment Card Industry Data Security Standard):** A mandatory security standard for organizations that handle branded credit cards, ensuring secure storage, transmission, and processing of cardholder data.
    - **Data Protection Regulations:** Legal frameworks (such as GDPR in Europe or local IT acts) governing consumer consent, data collection transparency, and breach notification requirements.

5. Digital Signatures and Public Key Infrastructure (PKI)

A. Concept of Digital Signatures

- **Definition:** A digital signature is a mathematical mechanism used to authenticate the identity of the sender of a digital message and ensure that the original content has not been altered in transit.
- **How It Works:**
    1. The sender passes the message through a **Cryptographic Hash Function** (like SHA-256) to generate a unique fixed-length hash digest.
    2. The sender encrypts this hash digest using their **Private Key**. This encrypted hash is the digital signature.
    3. The receiver decrypts the signature using the sender's **Public Key** to recover the hash digest.
    4. The receiver independently computes the hash of the received message. If both hashes match, authenticity, integrity, and non-repudiation are guaranteed.

B. Public Key Infrastructure (PKI) and Certificate Authorities (CAs)

- **Problem:** How do you know a public key truly belongs to the claimed entity (e.g., Amazon)?
- **Solution (PKI & CAs):** A **Certificate Authority (CA)** is a trusted third-party organization (e.g., Let's Encrypt, DigiCert) that issues digital certificates verifying the binding between a public key and its owner's identity.
- **Digital Certificate:** Contains the subject's public key, expiration date, serial number, and the CA's digital signature, establishing web trust.

E-COMMERCE & APPLICATIONS – UNIT IV

1. E-Commerce Payment Systems

An **Electronic Payment System (EPS)** is a financial settlement mechanism that enables the online transfer of funds between buyers and sellers over telecommunication networks, replacing physical cash, paper checks, or in-person card swipes.

Evolution of Online Payment Systems

- **1st Generation (Traditional Electronic Transfers):** Wire transfers and Automated Clearing House (ACH) networks used primarily for interbank transactions and payroll.
- **2nd Generation (Card-Based E-Commerce):** Integration of credit and debit card processing networks into web shopping carts using encrypted payment gateways.
- **3rd Generation (Digital Wallets & Real-Time Rail Networks):** Instant, account-to-account (A2A) settlement systems (e.g., UPI, Apple Pay, Google Pay) and platform-native wallets reducing intermediary fees and friction.

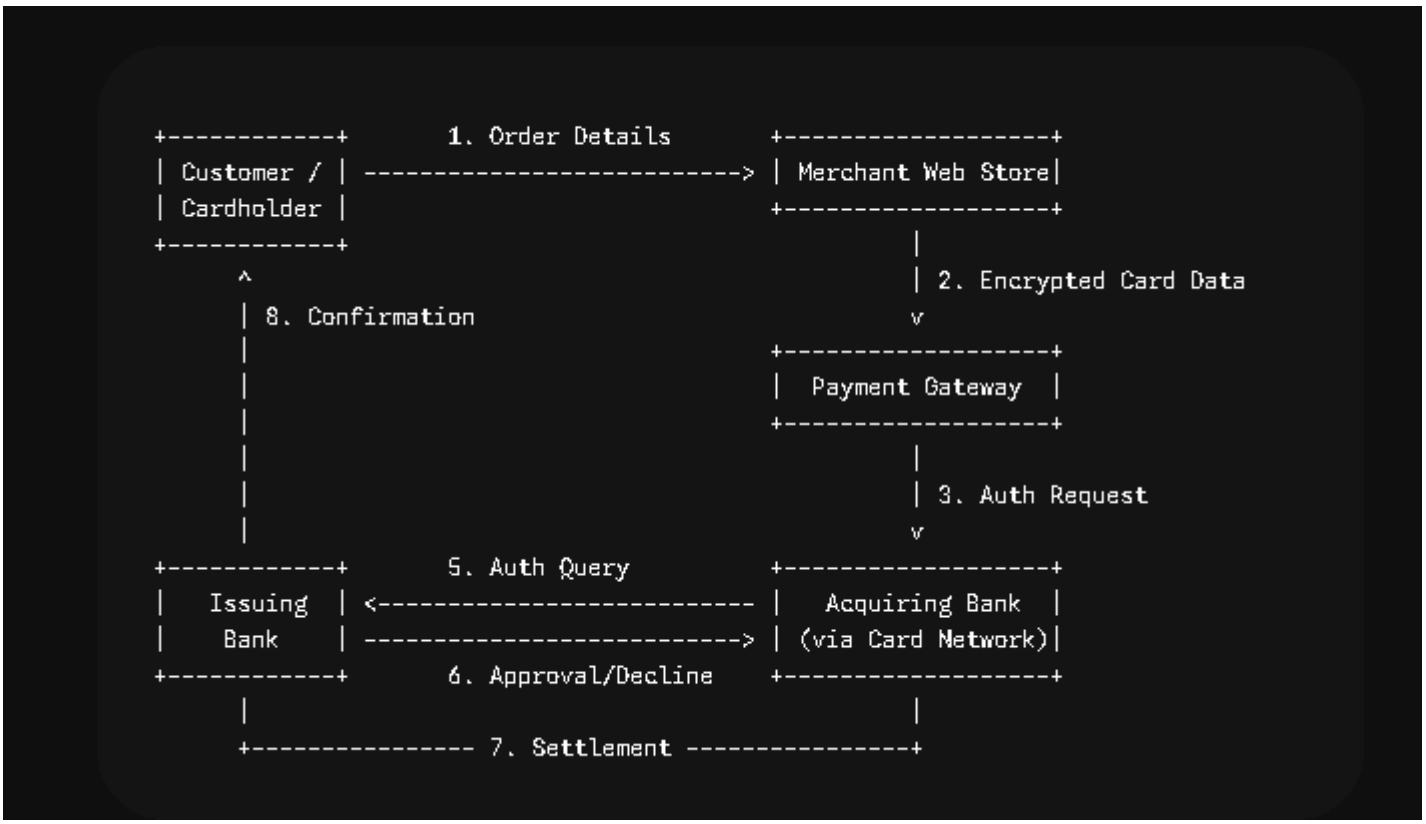
2. Credit Card E-Commerce Transactions

Credit card transactions remain one of the most common payment methods in global B2C e-commerce. Understanding the transaction lifecycle requires identifying the primary actors and processing stages involved.

Key Participants in a Credit Card Transaction:

1. **Cardholder:** The consumer purchasing goods/services online using a credit card.
2. **Merchant:** The online seller accepting credit card payments.
3. **Acquiring Bank (Merchant Bank):** The financial institution that holds the merchant's business account and processes credit card transactions on their behalf.
4. **Issuing Bank (Customer Bank):** The financial institution that issued the credit card to the customer and provides them with a credit line.
5. **Payment Gateway:** A specialized software service that securely encrypts credit card details entered on the merchant website and routes them between the merchant, acquirer, and payment networks.

6. **Payment Network (Card Association):** Organizations (e.g., Visa, Mastercard, American Express) that establish payment infrastructure, rules, and routing between acquiring and issuing banks.



Step-by-Step Transaction Flow:

1. **Initiation:** The customer places an order and enters credit card details (Card Number, Expiry Date, CVV) into an encrypted payment form.
2. **Authorization Request:** The **Payment Gateway** encrypts the data (using SSL/TLS) and sends it to the **Acquiring Bank**.
3. **Network Routing:** The Acquiring Bank routes the transaction through the **Payment Network** (e.g., Visa) to the customer's **Issuing Bank**.
4. **Verification & Hold:** The Issuing Bank verifies card validity, checks for fraud signatures, and ensures sufficient credit is available. If approved, it places a temporary hold on the funds and returns an **Authorization Code** back through the network to the merchant.
5. **Clearing and Settlement:** At the end of the business day, the merchant submits a batch of authorized transactions to their Acquiring Bank. The Acquiring Bank requests funds from the Issuing Bank through the card network.
6. **Payout:** The Issuing Bank transfers the funds to the Acquiring Bank (minus an interchange fee), and the Acquiring Bank deposits the net funds into the merchant's account.

3. B2C Digital Payment Systems

Modern B2C e-commerce utilizes diverse payment mechanisms beyond traditional credit cards to cater to varied user preferences and regional infrastructures:

Payment System	Working Mechanism	Primary Advantages	Key Examples

Debit Cards	Deducts funds directly from the customer's linked checking/savings bank account in real-time.	Low risk of consumer over-indebtedness; universally accessible.	Visa Debit, Mastercard Debit, RuPay
Digital Wallets / Mobile Pay	Stores payment credentials securely (often tokenized) in a digital application, allowing seamless one-touch checkout.	High speed, enhanced security via biometrics, zero exposure of underlying card details.	Apple Pay, Google Pay, PayPal
Account-to-Account (A2A) / Instant Payments	Directly links buyer and seller bank accounts via open banking APIs or real-time payment rails.	Extremely low transaction fees, immediate settlement, eliminates chargeback fraud.	UPI (India), Pix (Brazil), iDEAL (Netherlands)
Buy Now, Pay Later (BNPL)	Allows consumers to divide purchases into interest-free installments over time while paying the merchant upfront.	Increases average order value (AOV) and conversion rates for high-ticket items.	Klarna, Afterpay, Affirm
Stored-Value / Prepaid Cards	Funds are loaded onto a physical or virtual account prior to transactions.	Ideal for unbanked populations and strict budgeting control.	Gift cards, Reloadable virtual cards

4. B2B Payment Systems

Business-to-Business (B2B) payments differ significantly from B2C payments due to higher transaction values, complex purchasing workflows, credit terms, and administrative reconciliation needs.

Characteristics of B2B Payments:

- **Higher Order Values:** Transactions often range from thousands to millions of dollars.
- **Credit & Term-Based Invoicing:** Payments are rarely immediate; they rely on trade credit terms like **Net 30, Net 60, or Net 90** days.
- **Multi-Layered Approvals:** Requires purchase orders (POs), corporate invoice verification, and multi-department sign-offs.

Major B2B Payment Instruments:

1. **Automated Clearing House (ACH) & Electronic Funds Transfer (EFT):** Direct electronic batch transfers between corporate bank accounts, commonly used for recurring vendor payouts due to minimal transaction costs.
2. **Wire Transfers (SWIFT / Fedwire):** Real-time, high-value gross settlement systems used primarily for cross-border international trade transactions where immediate finality is required.
3. **Corporate / Purchasing Cards (P-Cards):** Specialized credit cards issued to corporate employees with preset spending limits, spending categories, and automated ERP integration to streamline procurement of indirect materials.
4. **Digital B2B Payment Networks & Escrow:** Online platforms that integrate e-invoicing, automated trade financing, and escrow services (releasing payment only when shipping milestones or quality inspections are met).

5. Privacy and Information Rights

A. The Privacy Challenge in E-Commerce

E-commerce platforms automatically harvest massive amounts of **Personally Identifiable Information (PII)**—including browsing history, purchase records, precise geolocation, device IDs, and demographic details.

- **Information Asymmetry:** Consumers often do not realize how extensively their behavioral data is profiled, aggregated, and monetized via third-party advertising networks.
- **Key Privacy Concerns:** Profiling, unauthorized data sharing, behavioral retargeting, dynamic pricing discrimination, and corporate data breaches.

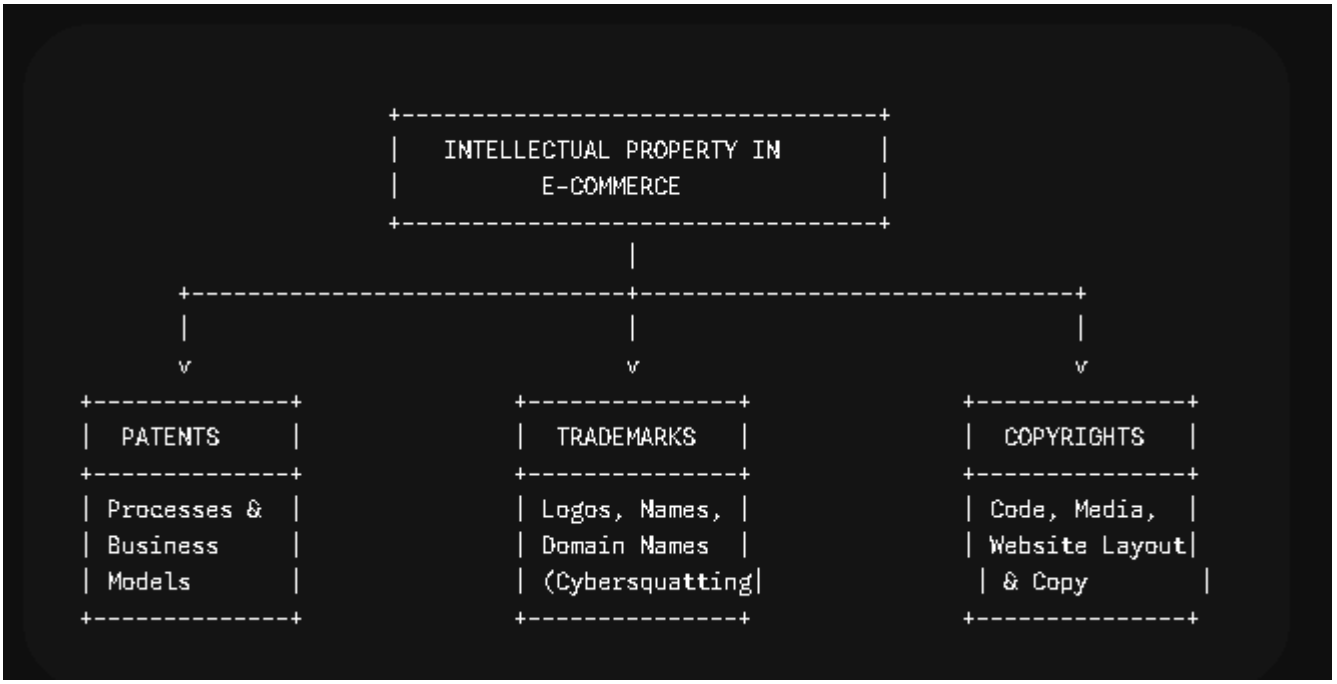
B. Core Principles of Information Rights (Fair Information Practices - FIP)

Regulatory frameworks worldwide base consumer information rights on fundamental principles:

1. **Notice / Awareness:** Merchants must explicitly state what data is collected, how it is used, and with whom it is shared via a clear **Privacy Policy**.
2. **Choice / Consent:** Consumers must be given options regarding how their information is used beyond the core transaction (**Opt-In** for sensitive data, **Opt-Out** for marketing).
3. **Access / Participation:** Consumers should have the right to view, correct, or request deletion of their personal data held by a firm (**Right to be Forgotten**).
4. **Integrity / Security:** Data collectors must implement reasonable security measures to prevent unauthorized data access or loss.
5. **Enforcement / Redress:** Mechanism to hold data handlers accountable for regulatory non-compliance.

6. Intellectual Property Rights (IPR) in E-Commerce

Intellectual property (IP) comprises intangible creations of the human intellect. In the digital environment, copying, distributing, and modifying IP carries near-zero marginal cost, making legal protection crucial.



Major Categories of IPR in E-Commerce:

1. Copyrights

- **Scope:** Protects original literary, dramatic, musical, artistic, and software works from unauthorized copying and distribution.
- **E-Commerce Application:** Protects website code (HTML, CSS, JavaScript), database architectures, product descriptions, marketing graphics, video content, and downloadable digital media.
- **Digital Enforcement: Digital Rights Management (DRM)** technologies and anti-circumvention laws prevent unauthorized copying or piracy of digital assets.

2. Trademarks

- **Scope:** Protects words, symbols, logos, phrases, or designs that identify and distinguish the source of goods or services of one brand from another.
- **E-Commerce Challenges:**
    - **Domain Name Disputes:** Registering internet domain names that mirror trademarked corporate brands.
    - **Cybersquatting:** Registering domain names corresponding to well-known trademarks in bad faith to resell them to the brand owner at exorbitant prices.
    - **Typosquatting:** Registering common misspellings of popular website URLs to redirect traffic to affiliate links or phishing sites.

3. Patents

- **Scope:** Grants an inventor exclusive legal rights to a novel, non-obvious, and useful invention, process, or machine for a limited time (typically 20 years).
- **E-Commerce Application:** Covers technological innovations, search engine algorithms, security encryption mechanisms, and **Business Method Patents** (e.g., Amazon's historical 1-Click checkout patent).